SmartBrew AI - Group 28 - Lab 3 report

SmartBrew AI is a local coffee-machine prototype for customers and maintenance technicians. It implements ten selected feature groups: accounts, menu, customization, personal recipes, availability, recommendations, checkout/history, loyalty, receipts and maintenance. An additional optional local language model interprets drink requests into customer-confirmed recipes. The complete elicitation retains camera/privacy requirements even though camera inference is outside this implementation.

Task A - Requirements elicitation, critical comparison and prompt record

Development models: Codex with student-confirmed selector label GPT-6.1 Sol (2026-10-20/21, initial Lab 3 Tasks A-C and web-chat handoff integration; exact backend build not exposed) and Gemini 4 Argon Medium through Google Antigravity (2026-10-21, local repository inspection, F11 multi-turn assistant evaluation/correction and F06 deterministic recommendation expansion and verification).

Runtime drink model: local Qwen3 4B (qwen3:4b) through Ollama 0.40.6. The development and runtime model roles are distinct.

The comparison uses the group's submitted Assignment 2 files, preserved unchanged with hashes in docs/baseline. Later review drafts are kept separately and do not replace the submitted baseline.

Ground the artifacts in the system description, feedback and course slides; use the six-column requirement template; retain the submitted baseline; trace capabilities to requirements, stories and backlog entries; distinguish source requirements from design assumptions; select ten demonstrable feature groups; implement and verify successful, rejected, ownership, retry and offline workflows. Refine the results through review and focused follow-up requests.

Main student prompt and selected key responses: output/G28Lab3_AI_Appendix.pdf and docs/AI_CONVERSATION.md. The appendix presents the prompting strategy and selected interactions; the original conversation is attached separately. Tasks A, B and C and subsequent F11/F06 updates use the recorded models; earlier Lab 1-2 tutoring and an intermediate web-chat Gemini draft before local IDE access did not expose confirmed model version labels.

User stories

ID	Name	Story	Requirements
US-01	Account registration	As a Customer, I want to create an account so that my preferences and purchase history belong to me.	FR-01
US-02	Customer login	As a Customer, I want to log in with credentials or my registered NFC card so that I can access my profile quickly.	FR-02
US-03	Preferences	As a Customer, I want to change my default drink settings so that future orders reflect my preferences.	FR-03
US-04	Menu and nutrition	As a Customer, I want to browse categories and nutrition so that I can choose a suitable available drink.	FR-04
US-05	Customization	As a Customer, I want to change milk, strength, temperature, syrup and espresso shots so that my drink matches my taste.	FR-05
US-06	Saved recipes	As a Customer, I want to name, save and modify my recipes so that I can reorder them with one touch.	FR-06
US-07	Availability	As a Customer, I want unavailable recipes blocked before purchase so that I do not pay for an impossible drink.	FR-07
US-08	Recommendations	As a Customer, I want available drinks ranked using my history, time and weather so that relevant choices are easy to find.	FR-08
US-09	Mood suggestions	As a Customer, I want optional mood-based suggestions so that I can receive a suitable recommendation without losing manual ordering.	FR-09/NFR-01
US-10	Natural-language ordering	As a Customer, I want the optional local LLM to suggest up to three recipes for my request so that I can compare them, select one and refine it before confirming a purchase.	FR-10
US-11	Payment	As a Customer, I want to pay with a supported card, wallet or employee balance so that I can complete a purchase.	FR-11
US-12	Order records	As a Customer, I want completed orders to retain the original settings so that I can inspect what I purchased.	FR-12
US-13	Loyalty	As a Customer, I want to earn and redeem loyalty points so that repeat purchases can reduce my future cost.	FR-13

ID	Name	Story	Requirements
US-14	Receipts	As a Customer, I want QR or email receipts so that I can keep proof of payment.	FR-14
US-15	Protected administration	As a Technician, I want secure administrative access so that customers cannot change maintenance state.	FR-15/NFR-02
US-16	Refills	As a Technician, I want to register refills so that ingredient and cup counters reflect full capacity.	FR-16
US-17	Waste safety	As a Technician, I want waste warnings and an overflow halt so that the machine stops safely until bins are emptied.	FR-17
US-18	Diagnostics	As a Technician, I want persistent local and authorized remote diagnostic logs so that faults can be investigated even after an outage.	FR-18
US-19	Service failure	As a Customer, I want manual ordering to remain available during service failure so that I can still buy with an offline-capable method.	FR-19
US-20	Offline status	As a Customer, I want clear offline payment status so that I know whether payment is approved or still pending.	FR-20
US-21	Image privacy	As a Customer, I want facial images discarded without storage or transmission so that mood analysis does not retain my face.	NFR-01
US-22	Reliable purchase	As a Customer, I want retries to create only one completed purchase so that I am not charged twice.	NFR-03
US-23	Responsive ordering	As a Customer, I want menu and quote responses to appear promptly so that choosing a drink remains convenient.	NFR-04
US-24	Repeatable assessment	As an Evaluator, I want a documented setup and feature walkthrough so that I can run the prototype on an independent machine.	NFR-05

Functional requirements - lecture template

The six columns follow Lecture 2 slide 47. Functional and non-functional items are labeled separately in Lab 3. For grouped Lab 2 items, labeled conditions apply to the corresponding operation.

ID	Description	Inputs	Outputs	Precondition	Postcondition
FR-01	Create an account The system shall register a customer account with unique credentials and a preference profile.	Email, password, display name.	Account identifier or validation error.	Email is not already registered; required values are valid.	One account/profile is stored; invalid registration makes no change.
FR-02	Authenticate customers The system shall authenticate credentials or a registered NFC/card identifier and load only the corresponding customer's profile.	Credentials or NFC identifier.	Authenticated session/profile or denial.	An account exists; supplied credential is registered.	Successful login creates a session; failed login grants no access.
FR-03	Maintain preferences The system shall let authenticated customers update their default milk, strength and temperature.	Customer session; preference values.	Saved preference profile.	Customer is authenticated; settings are supported.	Updated defaults persist across logout/restart.
FR-04	Browse menu and nutrition The system shall display beverage categories, ingredients, allergens, nutrition, prices and availability.	Category/filter; menu and stock data.	Matching drinks and nutritional details.	Local menu is available.	Customer can inspect drinks; unavailable items cannot be ordered.
FR-05	Customize a beverage The system shall validate milk, coffee strength, temperature, syrup and extra-shot choices and update the recipe, price and nutrition before confirmation.	Base drink and supported customization choices.	Validated recipe, price, nutrition or error.	A base drink has been selected.	The exact validated settings are visible; invalid options cannot be confirmed.
FR-06	Manage named recipes The system shall save, rename, modify and delete personal recipes and restore their settings for one-touch ordering, subject to current stock checks.	Customer session, recipe name/settings.	Stored recipe or restored order draft.	Customer is authenticated; recipe settings are valid.	Only the owner can change a recipe; later orders recheck availability.

ID	Description	Inputs	Outputs	Precondition	Postcondition
FR-07	Check ingredient availability The system shall reject menu selections and confirmed recipes if an ingredient is depleted, below its minimum threshold or insufficient for the requested serving.	Recipe quantities; stock levels/thresholds.	Availability result and missing-ingredient reasons.	Recipe and current counters exist.	Unavailable recipes cannot be purchased; stock is revalidated at checkout.
FR-08	Rank recommendations The system shall score available drinks using preferences, purchase history, time and weather, then reorder and highlight the best recommendations.	Profile/history, time, weather, available menu.	Ranked available drinks and recommendation reasons.	Local menu exists; history/profile may be empty.	Only available drinks are recommended; missing weather falls back to local context.
FR-09	Analyze mood The system shall use the embedded camera to infer a mood and optionally add it to the recommendation context; unavailable inference shall not prevent manual ordering.	Consented transient camera frame.	Mood label or unavailable result.	Camera/mood module enabled and customer opts in (design assumption).	Recommendations can use the label; no image is retained.
FR-10	Interpret natural-language requests When the optional local LLM is enabled, the system shall translate a broad drink request into up to three distinct supported recipe choices with short reasons and server-calculated prices; precise requests may return fewer choices. The customer selects a recipe before checkout, and follow-up changes use the selected recipe.	Request text; local model; supported menu/options; active selected recipe.	One to three validated recipe choices, a clarification question or an unavailable explanation.	Optional module is installed and enabled.	Only customer-confirmed and validated parameters reach ordering; proposing/applying recipes does not purchase. No technical scores are shown in the assistant.
FR-11	Process supported payments The system shall support EMV chip/contactless credit/debit, Apple Pay, Google Pay and prepaid employee cards and distinguish approved, declined and pending outcomes.	Confirmed recipe; method; provider response/prepaid balance.	Payment result and transaction reference.	Recipe is available; machine is not halted.	Only approved payment completes an order; declines change no stock, points or balance.

ID	Description	Inputs	Outputs	Precondition	Postcondition
FR-12	Record completed transactions The system shall record each completed purchase once with owner/guest reference, timestamp, original recipe settings, price and payment outcome and show the owner's history.	Approved purchase; recipe snapshot; idempotency key.	Persistent completed transaction/history entry.	Payment approved; checkout validations still hold.	Receipt/history preserve the purchased settings even after a saved recipe is edited.
FR-13	Earn and redeem loyalty The system shall let logged-in customers redeem earned points at checkout and award points after a completed paid purchase, with no changes on decline.	Customer balance; requested points; order total.	Discount, paid total and updated points.	Logged-in customer has enough points; redemption does not exceed price.	Spend/earn is committed once with purchase; a rejected purchase leaves points unchanged.
FR-14	Deliver receipts After a completed purchase, the system shall offer a QR receipt or email delivery and offer QR as fallback on email failure.	Purchase reference; selected channel; email if needed.	Receipt QR/link or email-delivery result.	A completed transaction exists; recipient choice is valid.	Receipt reflects the immutable transaction; a failed email does not lose the purchase.
FR-15	Authenticate maintenance personnel The system shall restrict administrative actions to technicians authenticated by physical key, registered master NFC card or technician PIN.	Technician PIN/master card/key signal.	Admin session or access denial.	Credential/key is registered for technician role.	Customers cannot refill ingredients, empty bins or read administrative diagnostics.
FR-16	Register refills The system shall allow authenticated technicians to register full ingredient/cup refills and reset the selected counter to its configured maximum.	Admin session; selected ingredient; refill confirmation.	Updated counter and audit event.	Technician has performed/confirmed the physical refill.	Counter equals maximum; unrelated stock is unchanged; availability is recalculated.
FR-17	Monitor waste and halt safely The system shall monitor liquid and grounds bins, warn near capacity and halt dispensing on full/overflow status until a technician records emptying.	Waste sensor/counter readings; emptying confirmation.	Warning/halt status and event log.	Bin capacities and current readings are known.	A full bin blocks ordering; authorized emptying resets that bin and re-evaluates the halt.

ID	Description	Inputs	Outputs	Precondition	Postcondition
FR-18	Diagnose errors The system shall show real-time faults on the admin dashboard and persist diagnostic events for authorized remote diagnostics when connectivity is available.	Fault code, message, severity, timestamp; diagnostic request.	Dashboard event and diagnostic log/export.	Fault reporting is active; diagnostics access is authorized.	Events remain available locally during a remote outage.
FR-19	Keep manual ordering available When network, weather or optional AI services fail, the system shall continue basic local menu/customization subject to ingredient, payment and safety rules.	Unavailable service; local menu/profile/stock.	Manual ordering interface and fallback indication.	Local machine hardware and required supplies are operational.	External-service loss alone does not block local selection/customization.
FR-20	Handle offline payment honestly The system shall use an approved offline-capable method or securely queue permitted pending payments and reconcile once on reconnect; it shall never present an unapproved online-only wallet payment as successful.	Connectivity; method policy; payment reference.	Offline approval, pending entry or denial; reconciliation result.	Payment method supports the chosen offline policy.	Pending authorizations do not dispense; approved prepaid orders and queued diagnostics persist.

Non-functional requirements - lecture template

ID	Description	Inputs	Outputs	Precondition	Postcondition
NFR-01	Protect facial-image privacy All camera frames shall be processed only in volatile memory, discarded immediately after inference and never written to disk, logs, databases or external servers.	Transient camera frame and inference lifecycle.	Mood label without image retention/transmission.	Any camera-based inference runs.	No facial-image bytes remain in persistent storage or outbound traffic.
NFR-02	Enforce access control The system shall check role/ownership on each protected request, hash passwords and technician PINs with per-credential salts, and never store raw payment card details.	Session; requested action; credentials.	Authorized response or denial; protected credentials.	A protected operation or credential storage is requested.	Other customers/admin data cannot be modified through forged requests.
NFR-03	Commit purchases consistently The system shall commit payment outcome, transaction snapshot, stock, waste, prepaid balance and loyalty changes atomically and idempotently.	Checkout/retry; current persistent state.	One consistent success or no completed purchase effects.	Checkout starts with a unique request key.	Retries cannot double charge/dispense; failure leaves no partial purchase.
NFR-04	Respond promptly Proposed acceptance target: local menu and quote responses shall complete within 2 seconds in at least 95 of 100 sequential requests on the documented demo machine, excluding external services.	100 local menu/quote requests.	Recorded durations and pass/fail count.	Demo environment is documented; server warmed up.	Measured results meet the target or the limitation is reported.
NFR-05	Provide reproducible setup The prototype shall run on a clean Python 3.12 environment with a modern browser using pinned runtime dependencies and documented setup, test and launch commands.	Source checkout; Python 3.12; dependency file.	Running prototype and repeatable test results.	Evaluator follows documented installation steps.	All ten selected features have executable demo steps; simulated integrations are identified.

Initial product backlog

Importance: proposed 0-100 ordering score, higher first. Estimate: relative story points, not hours. Initial planning values require group review. Description supplements the slide's fields; Notes provides traceability. No sprint commitment is made without team capacity.

ID	Name	Importance	Estimate	Description	How to demo	Notes
PB-FR-01	Create an account	100	3	The system shall register a customer account with unique credentials and a preference profile.	Register a new email; retry the same email and verify rejection.	FR-01; S01; F01
PB-FR-02	Authenticate customers	100	5	The system shall authenticate credentials or a registered NFC/card identifier and load only the corresponding customer's profile.	Use password and DEMO-NFC-001; wrong password is denied.	FR-02; S01; F01
PB-FR-07	Check ingredient availability	100	5	The system shall reject menu selections and confirmed recipes if an ingredient is depleted, below its minimum threshold or insufficient for the requested serving.	Refill then exhaust milk; verify milk drink blocked and black coffee still available.	FR-07; S08; F05
PB-FR-11	Process supported payments	100	8	The system shall support EMV chip/contactless credit/debit, Apple Pay, Google Pay and prepaid employee cards and distinguish approved, declined and pending outcomes.	Run each method using the simulator; decline one; prepaid insufficient funds is denied.	FR-11; S09; F07
PB-FR-12	Record completed transactions	100	5	The system shall record each completed purchase once with owner/guest reference, timestamp, original recipe settings, price and payment outcome and show the owner's history.	Buy an oat recipe, edit it to dairy and verify old history remains oat; retry checkout key and verify one charge.	FR-12; S02; F07
PB-FR-15	Authenticate maintenance personnel	100	5	The system shall restrict administrative actions to technicians authenticated by physical key, registered master NFC card or technician PIN.	Technician PIN creates admin session; customer token on maintenance endpoint is denied.	FR-15; S12; F10

ID	Name	Importance	Estimate	Description	How to demo	Notes
PB-FR-17	Monitor waste and halt safely	100	8	The system shall monitor liquid and grounds bins, warn near capacity and halt dispensing on full/overflow status until a technician records emptying.	Simulate full grounds, verify checkout blocked, empty grounds and verify ordering resumes.	FR-17; S14; F10
PB-NFR-01	Protect facial-image privacy	100	5	All camera frames shall be processed only in volatile memory, discarded immediately after inference and never written to disk, logs, databases or external servers.	Inspect filesystem/database and capture network traffic during a real inference run.	NFR-01; S16; Deferred
PB-NFR-02	Enforce access control	100	5	The system shall check role/ownership on each protected request, hash passwords and technician PINs with per-credential salts, and never store raw payment card details.	Call admin endpoints as customer; edit another user's recipe; inspect credential storage.	NFR-02; S01/S09/S12; security refinement; F01/F10
PB-NFR-03	Commit purchases consistently	100	8	The system shall commit payment outcome, transaction snapshot, stock, waste, prepaid balance and loyalty changes atomically and idempotently.	Retry same checkout; force decline/full bin and compare all counters before/after.	NFR-03; S02/S08/S09/G01; integrity refinement; F05/F07/F08
PB-FR-04	Browse menu and nutrition	95	3	The system shall display beverage categories, ingredients, allergens, nutrition, prices and availability.	Filter Coffee and Chocolate; inspect calories/allergens and availability.	FR-04; S07/S08; F02
PB-FR-05	Customize a beverage	95	5	The system shall validate milk, coffee strength, temperature, syrup and extra-shot choices and update the recipe, price and nutrition before confirmation.	Choose Latte/oat/strong/75 C/two syrup shots; verify quote and invalid value rejection.	FR-05; S07; F03
PB-FR-16	Register refills	90	3	The system shall allow authenticated technicians to register full ingredient/cup refills and reset the selected counter to its configured maximum.	Refill milk and cups; verify exact capacity and audit event.	FR-16; S13; F10

ID	Name	Importance	Estimate	Description	How to demo	Notes
PB-FR-19	Keep manual ordering available	90	5	When network, weather or optional AI services fail, the system shall continue basic local menu/customization subject to ingredient, payment and safety rules.	Disable network/weather; browse/customize and buy using prepaid balance.	FR-19; S17; F06/F07
PB-NFR-05	Provide reproducible setup	90	3	The prototype shall run on a clean Python 3.12 environment with a modern browser using pinned runtime dependencies and documented setup, test and launch commands.	Install pinned dependency, run unit tests and follow the ten-feature walkthrough.	NFR-05; Lab3 Task B/C; F01-F10
PB-FR-03	Maintain preferences	85	3	The system shall let authenticated customers update their default milk, strength and temperature.	Save oat milk/strong/75 C, log out/in and verify retained settings.	FR-03; S01; F01
PB-FR-20	Handle offline payment honestly	85	8	The system shall use an approved offline-capable method or securely queue permitted pending payments and reconcile once on reconnect; it shall never present an unapproved online-only wallet payment as successful.	Offline prepaid succeeds; wallet stays pending without stock/points changes; reconnect reconciles once.	FR-20; S17; F07
PB-FR-06	Manage named recipes	80	5	The system shall save, rename, modify and delete personal recipes and restore their settings for one-touch ordering, subject to current stock checks.	Save Morning Oat, rename/edit it and order it again; another user cannot edit it.	FR-06; S03; F04
PB-FR-18	Diagnose errors	80	5	The system shall show real-time faults on the admin dashboard and persist diagnostic events for authorized remote diagnostics when connectivity is available.	Inject a grinder jam; verify time/severity/local log and authorized JSON export.	FR-18; S15; F10

ID	Name	Importance	Estimate	Description	How to demo	Notes
PB-FR-08	Rank recommendations	75	8	The system shall score available drinks using preferences, purchase history, time and weather, then reorder and highlight the best recommendations.	Switch period mode (local clock or manual morning/afternoon/evening/night), temperature (cold/mild/warm/hot) and weather condition (unknown/sunny/cloudy/rainy/snowy); buy a drink and verify ranking and reasons.	FR-08; S04/S06/S17; F06
PB-FR-13	Earn and redeem loyalty	70	5	The system shall let logged-in customers redeem earned points at checkout and award points after a completed paid purchase, with no changes on decline.	Redeem 100 points, inspect discount/new balance; repeat with declined payment and verify no deduction.	FR-13; G01; F08
PB-FR-14	Deliver receipts	70	5	After a completed purchase, the system shall offer a QR receipt or email delivery and offer QR as fallback on email failure.	Open a real QR; request email in local outbox; fail email and verify QR fallback.	FR-14; S10; F09
PB-FR-09	Analyze mood	45	8	The system shall use the embedded camera to infer a mood and optionally add it to the recommendation context; unavailable inference shall not prevent manual ordering.	With real module enabled, infer tired mood; deny consent and verify manual ordering still works.	FR-09; S05/S16; Deferred
PB-NFR-04	Respond promptly	45	3	Proposed acceptance target: local menu and quote responses shall complete within 2 seconds in at least 95 of 100 sequential requests on the documented demo machine, excluding external services.	Measure 100 requests; report p95 and environment without claiming production SLA.	NFR-04; Design assumption; not in brief; F02/F03

ID	Name	Importance	Estimate	Description	How to demo	Notes
PB-FR-10	Interpret natural-language requests	30	13	When the optional local LLM is enabled, the system shall translate a broad drink request into up to three distinct supported recipe choices with short reasons and server-calculated prices; precise requests may return fewer choices. The customer selects a recipe before checkout, and follow-up changes use the selected recipe.	Request three strong oat options; compare names, settings, short reasons and prices; apply option two and refine its strength. Verify no automatic purchase, unsupported-option refusal and manual fallback. Keep F06 scoring rationale in technical documentation, not assistant cards.	FR-10; S11; approved three-choice UI refinement; F11 (optional)

Comparison with the submitted Lab 2

The comparison uses the group's submitted Assignment 2 files, preserved unchanged with hashes in docs/baseline. Later review drafts are kept separately and do not replace the submitted baseline.

Topic	Submitted Lab 2	LLM-assisted revision	Decision and reason
Requirement format	13 short ID/name/shall rows; no inputs, outputs or conditions.	All FR/NFR rows use the six lecture fields; failure postconditions are explicit.	Accept: directly addresses the negative feedback.
REQ-01 / menu	Categories and nutrition specified.	FR-04 keeps scope; FR-08 explicitly reorders/highlights scored recommendations.	Accept: menu scoring behavior is in the system description.
REQ-02/03 / accounts	Login/profile; creation is ambiguous; saved recipes absent.	FR-01/02/03 separate registration, auth and defaults; FR-06 names/modifies/restores recipes.	Accept: account and recipe requirements were underrepresented.
REQ-04/05 / drink	Customization and threshold checks present.	FR-05 validates options and quote; FR-07 also checks serving quantity and rechecks at purchase.	Accept: prevents invalid or stale-stock orders.
REQ-06/07 / AI	Context recommendations and in-memory mood present; consent and fallbacks unclear.	FR-08/09 plus NFR-01 preserve mood privacy and provide explicit opt-in/failure handling.	Accept source behavior; opt-in is a design assumption requiring group review.
REQ-08 / LLM	Local LLM required as if always installed.	FR-10 explicitly optional/modular and validates customer-confirmed output.	Accept: optional status matches the brief. F11 now supplies real local inference beyond the ten core groups.
REQ-09 / payment	Contactless, wallets, employee card; chip and offline flow absent.	FR-11 adds EMV chip; FR-20 distinguishes offline prepaid, pending wallet and once-only reconciliation.	Accept coverage; pending/non-dispensing is an explicit conservative demo policy.
Transaction recording	No explicit requirement for immutable customization history.	FR-12 and NFR-03 record snapshots atomically and prevent duplicate purchase effects.	Accept: S02 requires every completed transaction to be recorded.
REQ-10 / loyalty	Redemption only; how points are earned is undefined.	FR-13 defines earning after success, bounded redemption and no deduction on failure.	Accept with documented demo conversion rules; group must review those rules.

Topic	Submitted Lab 2	LLM-assisted revision	Decision and reason
REQ-11 / receipt	QR/email included.	FR-14 adds delivery failure/QR fallback and immutable receipt source.	Accept: concrete observable behavior for existing story.
REQ-12 / maintenance	Secure admin/refills/waste monitoring stated together.	FR-15/16/17 identify credentials, exact counter reset, alerts and overflow halt.	Accept: explicit safety behavior was missing.
REQ-13 / diagnostics	Store/display real-time faults for diagnostics.	FR-18 adds authorized remote diagnostic availability and local preservation during outage.	Accept: matches source; prototype supplies export, not a deployed remote service.
Non-functional requirements	Privacy mixed into mood; no separate integrity/setup/performance criteria.	NFR-01..05 define privacy, security, atomicity, proposed response target and reproducible setup.	Accept privacy/security/integrity; 2-second target is proposed, not a source mandate.
Backlog	13 rows with accepted demos and estimates, no explicit trace links/units.	25 rows use same required columns plus notes, FR/NFR/story links and story-point units.	Accept: preserves artifact structure; estimates remain planning judgments.

User-story comparison: original to revised

Story IDs are local to each assignment: Lab 2 US-04 and Lab 3 US-04 describe different goals. The left column and quoted wording below come from docs/baseline/submitted-task2b.md, not the later generated Lab 2 review reference. All 13 submitted stories are retained, clarified or split into 17 revised stories; seven additional stories bring the total to 24. No original customer/technician goal is silently dropped. More stories indicate finer granularity and restored coverage, not proof of higher quality or implementation of deferred capabilities.

Submitted story	Original wording	Revised story links	Change and justification
Lab 2 US-01 / Browse Menu and Nutrition	As a Customer, I want to browse beverage categories and inspect nutrition information so that I can choose a drink that fits my preferences and dietary needs.	Lab 3 US-04 / Menu and nutrition	Retain: the customer goal is unchanged; availability is explicit. Illustrative nutrition does not establish dietary or allergy safety.
Lab 2 US-02 / Log In	As a Customer, I want to log in using credentials or an NFC/card tap so that I can quickly access my personal preference profile.	Lab 3 US-02 / Customer login	Retain and clarify: the card must be registered to the corresponding account. Explicit authentication and ownership checks address the observed Alex-login defect.
Lab 2 US-03 / Create or Manage Profile	As a Customer, I want to create and manage my preference profile so that the machine can remember my favorite drink settings.	Lab 3 US-01 / Account registration; US-03 / Preferences	Split: account creation and editing defaults have different authentication conditions and independently observable results. Saved recipes are separately added as US-06.
Lab 2 US-04 / Customize Beverage	As a Customer, I want to customize beverage parameters such as milk type, coffee strength, temperature, and syrup shots so that I receive the exact drink I want.	Lab 3 US-05 / Customization	Clarify: explicitly include extra espresso shots and supported settings; server validation and a visible quote make the existing goal testable.
Lab 2 US-05 / Check Ingredient Availability	As a Customer, I want the machine to prevent unavailable drink selections so that I do not order a beverage that cannot be prepared.	Lab 3 US-07 / Availability	Retain and strengthen: reject unavailable customized recipes before payment. Threshold, serving quantity and checkout revalidation are specified in FR-07.
Lab 2 US-06 / Generate Recommendations	As a Customer, I want to receive personalized beverage recommendations so that I can quickly choose a drink based on my preferences, history, time of day, and context.	Lab 3 US-08 / Recommendations	Clarify: identify weather as context and rank available drinks. Preferences remain an explicit input in FR-08. Service-failure behavior is separately added as US-19.
Lab 2 US-07 / Analyze Mood	As a Customer, I want the machine to analyze my mood when recommendation mode is used so that it can suggest drinks that match my current state.	Lab 3 US-09 / Mood suggestions; US-21 / Image privacy	Separate recommendation benefit from image protection. Opt-in is a proposed design refinement; in-memory disposal is a source constraint. Camera implementation remains deferred.

Submitted story	Original wording	Revised story links	Change and justification
Lab 2 US-08 / Process Natural Language Order	As a Customer, I want to request a drink using free-form natural language so that I can order complex custom drinks without manually navigating every menu option.	Lab 3 US-10 / Natural-language ordering	Correct scope: the model is optional and proposes supported settings for confirmation. A later approved UI refinement offers up to three recipes with reasons/prices and selected-recipe follow-ups, without technical scores. Reject treating a proposal as permission to purchase.
Lab 2 US-09 / Process Payment	As a Customer, I want to pay using supported methods such as contactless card, digital wallet, or prepaid employee card so that I can complete checkout quickly.	Lab 3 US-11 / Payment	Retain: supported payment choice remains the customer goal. FR-11 specifies methods and outcomes; US-12, US-20 and US-22 separately expose recording, offline status and retry integrity.
Lab 2 US-10 / Redeem Loyalty Points	As a Customer, I want to redeem loyalty points earned from previous purchases so that I can receive discounts or free drinks.	Lab 3 US-13 / Loyalty	Expand: earning becomes explicit alongside redemption. Accept the group's extension; conversion rates are documented demo assumptions, not source requirements.
Lab 2 US-11 / Receive Receipt	As a Customer, I want to receive a receipt by QR code or email so that I can keep proof of my purchase.	Lab 3 US-14 / Receipts	Retain: the two channels and benefit are unchanged. FR-14 adds QR fallback when email fails; prototype email delivery is simulated.
Lab 2 US-12 / Manage Inventory	As a Technician, I want to register ingredient refills and monitor waste levels so that the machine remains operational and does not offer drinks with depleted ingredients.	Lab 3 US-15 / Protected administration; US-16 / Refills; US-17 / Waste safety	Split: authorization, full-counter refill and overflow protection are different outcomes. Accept explicit cup refills, warnings and halt behavior from the system description.
Lab 2 US-13 / View Error Logs	As a Technician, I want to view real-time machine error logs so that I can diagnose problems such as grinder jams, heater faults, and low ingredient thresholds.	Lab 3 US-18 / Diagnostics	Expand: include persistence through outages and authorized remote diagnosis from S15/S17. The prototype supplies local logs/export; remote deployment remains deferred.

Seven additional stories and baseline provenance

New story	Requirement / source	Reason for addition
US-06 / Saved recipes	FR-06; S03	Add: the submitted stories omitted naming, modifying and restoring personal recipes. Preferences alone do not cover reusable named recipes.
US-12 / Order records	FR-12; S02	Add: record every completed transaction with its original customization settings. Payment alone does not express later history inspection.
US-19 / Service failure	FR-19; S17	Add: preserve manual ordering when network or optional services fail, subject to payment, stock and safety rules.
US-20 / Offline status	FR-20; S17	Add: distinguish pending online-only authorization from a completed offline-capable purchase. Pending without dispensing is the documented prototype policy.
US-22 / Reliable purchase	NFR-03; design refinement of S02/S09/S17	Add: make once-only retry and consistent purchase effects observable. Atomicity/idempotency are engineering refinements, not verbatim source wording.
US-23 / Responsive ordering	NFR-04; proposed design assumption	Add: make responsiveness measurable. The two-second target is proposed and applies only to warmed local menu/quote calls.
US-24 / Repeatable assessment	NFR-05; Lab 3 Tasks B/C	Add: make independent-machine setup and feature demonstration explicit. This is an assessment need rather than a customer feature in the system description.

Comparison baseline: the original group submission is the reference for the before-and-after analysis. Each original story is mapped to its revised counterpart, and new stories have separate justifications. The comparison evaluates coverage, clarity, traceability and the fit with the system description.

Rejected alternatives and critical evaluation

These alternatives were evaluated against the system description and selected prototype scope.

Option	Justification
Cloud facial-image analysis or image retention	Reject: directly contradicts the mandatory in-memory/no-transmission constraint.
Force account login before any manual order	Reject: account-only ordering is not required and would weaken basic manual fallback.
Mark offline Apple Pay/Google Pay as approved without processor support	Reject: the prototype cannot assume external financial approval. Show pending and do not dispense.
Call a keyword parser a local LLM or call a mood dropdown computer vision	Reject: would misrepresent implementation. F11 uses actual local-model inference; camera/mood inference remains deferred.
Subscriptions, social sharing and multiple-drink baskets	Reject from scope: not requested; no need to enlarge the minimum usable prototype.
Conflate submitted Assignment 2 with later AI-generated review corrections	Reject: use the preserved group submission for a consistent before-and-after comparison. Substituting later review corrections would obscure the original differences.

Initial name-only model checks missed requirement fields and wrong/empty use-case descriptions. Later, the student found a browser login defect that passing tests had missed: a hidden NFC value selected Alex instead of checking the entered password. The new regressions verify separate identities and ownership. Real-model testing also found an invented usual-drink recipe; the failed output was retained and clarification rules were refined. Subsequent Google Antigravity (Gemini 4 Argon Medium) evaluation and implementation resolved multi-turn F11 context/constraint defects (8/8 in evidence/current/assistant-heldout-postchange.json) and expanded F06 recommendation scoring and verification. The subsequent three-choice assistant revision passes 38 unit tests (evidence/current/unit-tests.txt) and 65 browser checks (evidence/current/browser-results.json), including the reproduced selected-option strength correction. These observations distinguish observed evidence from an unsupported claim of completeness.

Cross-lab continuity and revisions

Lab 1 received a positive grade, with Partial feedback on component scope and UI dependency on the AIEngine. Its nine domain concepts map to roles, preferences, recipes, purchase records and stock/waste state, not necessarily one code class each. The optional local-LLM responsibility is now implemented through a separate adapter; camera/mood inference remains deferred. Loyalty first appears in the supplied Lab 2, not the Lab 1 diagrams. See docs/CROSS_LAB_REVIEW.md for the detailed source mapping.

Read-only Lab 1 inspection preserved the original model and hashes. The visible User {Abstract} label is not backed by Astah's native abstract flag, although the grader accepted the class criteria. No original model is silently rewritten. Conceptual model mappings and implementation scope differences are documented in the cross-lab review.

Reliability follow-up: HTTP 503 from recommendations reproduced a blank menu in the isolated browser test. Essential ordering now renders before optional ranking, whose wait is limited to 1.2 seconds. Manual browsing, customization and payment retain their standard-menu path. This is an ordinary regression fix, not the unselected optional student experiment.

Customer checkout no longer controls its payment result. Only the technician's Test scenarios area changes the persistent external-processor mode. Normal sign-in fields start empty, mixed authentication requests are rejected, and account changes clear private UI and assistant conversation. Existing data is preserved; historical orders are not automatically reassigned.

Course policy: the earlier generated Lab 2 corrections remain review aids only, not submission materials. Original Lab 1-2 sources are unchanged. Lab 3 permits generation; the separate appendix records assistance and source provenance.

The instructor strongly recommends additional experimentation, but does not state a fourth mandatory task. During Lab 3 verification, the project conducted an empirical before/after evaluation of multi-turn local-model behavior and deterministic recommendation ranking against isolated temporary databases: the 8-turn held-out assistant baseline (evidence/history/assistant-heldout-baseline.json, 5/8 turns passed) exposed a leaked synthetic context turn, base-drink rigidity, silent substitution of unsupported iced/caramel requests and unrequested extra espresso shots; replacing the synthetic assistant turn with a trailing system context snapshot and ordering config before message improved results to 7/8 on the first post-change run (evidence/history/assistant-heldout-postchange-first-run.json) and 8/8 on the final regression run (evidence/current/assistant-heldout-postchange.json), alongside 12/12 in the conversation suite (evidence/current/assistant-conversations.json), 9/9 recipe turns in the long-chat suite (evidence/current/assistant-long-chat.json) and the current extended 65/65 browser checks (evidence/current/browser-results.json, with first-run browser regressions preserved in evidence/history/browser-failure.json and evidence/history/browser-f06-first-run-failure.json). The evaluation records retain both the initial failures and the final regression results.

Task B - Ten selected feature groups

ID	Functionality	Requirements	Demonstration
F01	Accounts, login and preferences	FR-01/02/03, NFR-02	Create account; password/NFC login; save default milk/strength/temperature; log out/in.
F02	Menu, categories and nutrition	FR-04	Browse six drinks; filter Coffee/Chocolate; inspect estimated nutrition and allergens.
F03	Validated custom drink quote	FR-05	Change milk, strength, temperature, syrup and shots; observe server-calculated quote.
F04	Named personal recipes	FR-06	Save Morning Oat; rename; load and edit settings; save update; delete.
F05	Ingredient/quantity checks	FR-07, NFR-03	Set oat level below threshold as technician; verify oat recipe cannot be purchased; refill.
F06	Context/history recommendations	FR-08/19	Toggle Recommended first; switch local-clock or manual period (morning/afternoon/evening/night), temperature (cold/mild/warm/hot) and condition (unknown/sunny/cloudy/rainy/snowy); verify history, saved-recipe and preference reasons and offline fallback.
F07	Checkout and immutable history	FR-11/12/20, NFR-03	Use five payment methods; set external decline in technician Test scenarios; inspect history/retries and offline reconciliation.
F08	Earn and redeem loyalty	FR-13	Apply 100 points; verify SEK 10 discount and success-only deduction/earning.
F09	QR receipt and email adapter	FR-14	Open receipt QR/text; send to local simulator outbox; fail email and retain QR fallback.
F10	Protected maintenance/diagnostics	FR-15/16/17/18	PIN/master-NFC login; full refill; waste halt/empty; fault logging; authorized diagnostic export.

Optional F11 - Local drink assistant

ID	Functionality	Requirements	Demonstration
F11	Local drink assistant	FR-10/19	Ask for three strong oat options; compare names, short descriptions, settings and server prices without technical scores. Apply option two, request a strength change and verify that recipe remains active. Precise requests may yield fewer choices; checkout remains separate.

One-time Windows setup: install Ollama and run tools/start_local_ai.ps1 to start it, download qwen3:4b if missing and warm the model. Later, tools/start_smartbrew.ps1 starts/reuses Ollama and launches SmartBrew together, using existing model weights without a download or warm-up wait. Missing AI leaves manual ordering available. Direct app.py launches require Ollama to be running separately. The model download is about 2.5 GB. The tested system has about 32 GB RAM and a 6 GB RTX 3060 Laptop GPU. Actual model digest, requests, responses and timings are in evidence/current/local-ai-results.json; these are selected examples, not a general accuracy guarantee.

The model returns a bounded recipe or clarification schema. The server rejects unsupported fields/options, recalculates price and checks current stock. Use recipe applies the proposal only after customer confirmation; checkout is separate. Inference cannot execute code or purchase. Missing/invalid/slow inference leaves manual ordering available. The cold-load wait is bounded at 90 seconds and excluded from menu/quote latency measurements.

Broad taste requests produce up to three distinct choices in one model response; precise named drinks, exact refinements or limited availability may produce fewer. Cards show names, short descriptions, settings and server prices, with no technical score or confidence percentage. The separate F06 scoring rationale stays in the architecture/backlog. Choosing any card makes it the active recipe for further requests. An observed wrong-drink response to 'Make it mild' is preserved; a narrow strength-only guard and real-model regression verify that this command changes only the selected recipe's strength. Other free-form interpretation remains fallible.

Implementation scope and assumptions

Prototype money uses integer ore (100 ore = SEK 1); demo prices/nutrition/ingredient quantities are illustrative and are not production dietary specifications.

Loyalty: earn one point per whole SEK actually paid; one point redeems SEK 0.10, bounded by balance and price. Demo Alex begins with 250 points and SEK 200 prepaid credit; new accounts begin with zero.

Customer orders may be placed as guest; profile/recipe/history and prepaid/loyalty features require customer login. Guest receipts use an unguessable capability token.

Mood opt-in and the proposed 2-second response target are design refinements; the original brief does not prescribe consent UI, a numerical latency target or point-conversion rules.

Weather/time, NFC/key signals, machine sensors, payment approval and email are simulated locally. Recommendation scoring uses separate deterministic affinity tables (preserving generic recipe defaults), period mode (manual technician override or injectable local clock across morning/afternoon/evening/night), online temperature (cold/mild/warm/hot), online weather condition (unknown/sunny/cloudy/rainy/snowy), and customer-owned preferences, saved recipes (+15) and completed-order count (+12 per order up to +36 as 'A frequent choice'). Manual ordering and stored-value prepaid payments work while the simulated network is offline.

Online-only payment requests can be persisted as pending without dispensing. Reconnect and customer Retry performs authorization/revalidation once; the prototype never assumes that a wallet can approve offline.

Camera mood inference remains deferred. F11 adds actual optional local Qwen3 4B inference through Ollama 0.40.6, beyond the ten selected groups. A later approved UI refinement offers up to three supported recipes with short descriptions and server prices, without technical scores. Selected-option context is preserved for further changes; exact named-strength follow-ups also have a narrow consistency guard. No proposal or application purchases or modifies accounts.

No physical brew command or real card authorization occurs. Admin diagnostics can be exported locally; there is no deployed remote diagnostic service or physical key reader.

Technician-only Test scenarios controls the persistent external payment mode (Normal/Decline). Customer checkout cannot submit an outcome override. Existing accounts/orders are retained; historical purchases are not reassigned after the authentication fix.

The model may need a cold load; server inference has a 90-second bound and manual ordering stays independent. The proposed menu/quote latency target excludes optional inference. Model proposals require customer review and fresh server validation.

Task C - Setup, repository and validation

Repository: https://github.com/kaantanidir/Software_Engi_1_LAB3

Submission: SUBMISSION.md maps the assessed artifacts; upload output/SmartBrew_Lab3_Canvas.zip to Canvas; Task B is referenced through the repository. Repository visibility was verified public on 26 October 2026. A ZIP-extracted source copy passed 38 unit tests in a fresh Python environment with the pinned dependencies and no workspace packages or personal data. A separate clean-install browser check verified three example prompts, real local inference, mobile layout and the reduced-motion loading indicator; the complete live-model browser suite passed 65 checks. Evidence: evidence/current/clean-install.txt and evidence/current/ux-polish.json. The repository contains the source and execution guide.

The development conversation is attached as docs/DEVELOPMENT_TRANSCRIPT.md: 251 recorded student/Codex messages with personal details redacted. Source hashes and transcript integrity information are in docs/DEVELOPMENT_TRANSCRIPT_MANIFEST.json. The AI appendix provides the prompting approach and selected interactions. The separate Gemini/Antigravity session is summarized in the interaction record; its original transcript was not available.

Runtime: Python 3.12 (verified 3.12.14), modern browser, pinned qrcode 8.2 (colorama 0.4.6 on Windows). SQLite and HTTP service use the Python standard library. Setup: python -m venv .venv; activate; python -m pip install -r requirements.txt. Windows run: powershell -NoProfile -ExecutionPolicy Bypass -File tools/start_smartbrew.ps1. This starts/reuses optional Ollama without downloading weights; -SkipAI omits AI preparation. Direct Python alternative: python app.py --port 8765. Visit http://127.0.0.1:8765. Tests: python -m unittest discover -s tests -v.

Demo customer: alex@smartbrew.test / DemoPass123!; registered NFC DEMO-NFC-001. Technician PIN 246810; master NFC MASTER-NFC-001. These are public simulator fixtures, not production credentials.

README.md contains all feature walkthroughs. docs/README.md indexes architecture, verification, prompt records and historical sources; older Lab 2 notes are in docs/reference/lab2. All authored functions document purpose/input/output. Runtime data, model weights and local packages are excluded from Git. See docs/QA_RESULTS.md and evidence/ for actual tests. Simulator approval is not financial or hardware validation.

References: provided feedback, system description and assignment instructions; Lecture 2 slides 20-23, 30-31, 38, 43, 47-48; Lecture 3 slides 17, 22, 28 and 38-40. Recommended reading: Henrik Kniberg, Scrum and XP from the Trenches (InfoQ).

Appendix - Source coverage

Source	Capability	Requirement(s)
S01	Accounts, credential/NFC login and personal preference profiles.	FR-01; FR-02; FR-03; NFR-02
S02	Record every completed transaction with beverage customization parameters.	FR-12; NFR-03
S03	Save, modify and name personal recipes for one-touch ordering.	FR-06
S04	Recommend using time, weather and historical choices.	FR-08
S05	Use embedded-camera mood analysis for tailored suggestions.	FR-09
S06	Reorder and highlight menu items using recommendation scores.	FR-08
S07	Browse categories, inspect nutrition and customize in real time.	FR-04; FR-05
S08	Prevent selection of drinks with depleted/below-threshold ingredients.	FR-04; FR-07; NFR-03
S09	Support credit/debit EMV chip/contactless, Apple Pay/Google Pay and prepaid employee cards.	FR-11; NFR-02; NFR-03
S10	Offer receipts by QR scan or email.	FR-14
S11	Optional modular local LLM for natural-language interaction and customization.	FR-10
S12	Secure technician mode using physical key, master NFC card or PIN.	FR-15; NFR-02
S13	Register ingredient/cup refills and reset counters to maximum capacity.	FR-16
S14	Monitor liquid/grounds bins, alert before disposal and halt on overflow.	FR-17
S15	Display errors on admin dashboard and log events for remote diagnostics.	FR-18
S16	Facial images: infer only in memory, discard immediately, never persist/transmit.	FR-09; NFR-01
S17	Network/service failure: manual ordering and supported offline payment/deferred processing.	FR-08; FR-19; FR-20
G01	Group feature: loyalty points earned on completed purchases and redeemed at checkout.	FR-13; NFR-03